

BharatTrip Refund Escalations: Business Case & Solution Proposal

Executive Summary (TL;DR)

BharatTrip is experiencing a spike in customer/agent refund escalations despite stable refund volumes. Our analysis identified a severe operational disconnect between Support and Finance, leading to **100 dropped tickets** and **149 short payments**. By implementing a **Single Source of Truth (SSOT)** and an **AI-Driven Data Reconciliation Workflow**, we deploy a "Digital Workforce." This allows existing teams to eliminate manual data leakage and automate agent communications, driving a projected **85%+ drop in escalations** over the next quarter **without adding a single headcount**.

1. Defining the Problem

Customer and agent escalations regarding refunds are rising. The core underlying problem is a **severe operational disconnect between the Support and Finance teams**, characterized by siloed tracking systems and inconsistent definitions of refund statuses.

Key Communication Failures:

1. **Data Leakage:** Informal requests via WhatsApp and Email bypass the tracker altogether, meaning they are never reviewed or processed.
2. **“Closed” Status Ambiguity:** The Support team marks tickets as “Closed” once handed off to Finance, misleading the agent into believing the refund has been disbursed.
3. **Hidden Policy Deductions:** Finance applies necessary deductions (e.g., cancellation fees) and processes payouts, but this information is never relayed back to the agent by Support, causing “short payment” escalations.

2. Data Analysis & Insights

By cross-referencing the `Support_Tracker`, `Finance_Tracker`, and `Escalations` logs via a deep database join, we identified the specific patterns driving the **155 total escalations**.

The Discrepancy Breakdown

Escalation Driver	Ticket Volume	Description
Dropped Handoffs	100	Logged by Support, but entirely missing from Finance tracking.
Deduction Mismatches	149	Amount promised by Support differs from amount disbursed by Finance.
Off-Tracker Leakage	24	Escalations with “no ref” originating from unlogged WhatsApp/Email.
Asynchronous Closure	47	Processed by Finance but missing from Support.

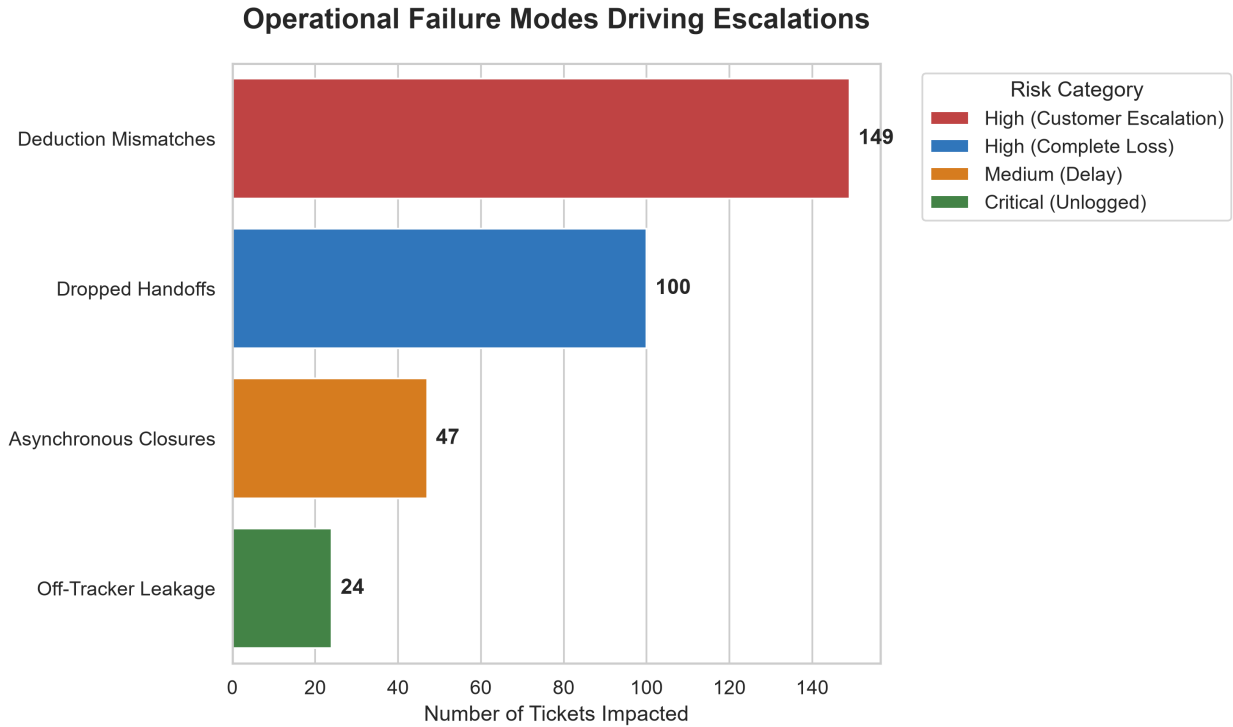


Figure 1: Visual breakdown of current support processing inefficiencies highlighting the 100 dropped handoffs and 149 mismatches.

Insight: While a naive net volume calculation (755 Support - 689 Finance) suggests only 66 missing tickets, our deep ID-to-ID database reconciliation reveals the true scale of the failure is **100 dropped tickets**, masked by duplicate records and asynchronous closures.

3. Designing a Zero-Headcount Solution

To permanently bring escalations down without adding headcount, BharatTrip must establish a unified **Single Source of Truth (SSOT)** and automate the communication loop.

The Proposed Workflow (Digital Workforce):

1. **Unified SSOT Tracking:** Migrate both teams onto a single shared database.
2. **AI-Powered Ingestion:** Deploy an AI agent to monitor Support Email and WhatsApp. The AI acts as a digital clerk, parsing unstructured messages and logging them as structured tickets, eliminating “no ref” leakage entirely.
3. **Automated Reconciliation:** A system trigger automatically cross-references Finance payouts with Support quotes. If a deduction occurs, the AI drafts an explanatory email for the agent, preventing short-payment shock.

4. The AI Prototype: Enterprise-Ready Architecture

To deliver a production-ready solution, I developed a **Streamlit Web Application** showcasing an advanced Multi-Agent architecture powered by **gemini-3.5-flash**. Rather than building a simple LLM script, I engineered a secure, deployable enterprise tool.

Fig 1: Proposed System Architecture for Multi-Agent Data Reconciliation

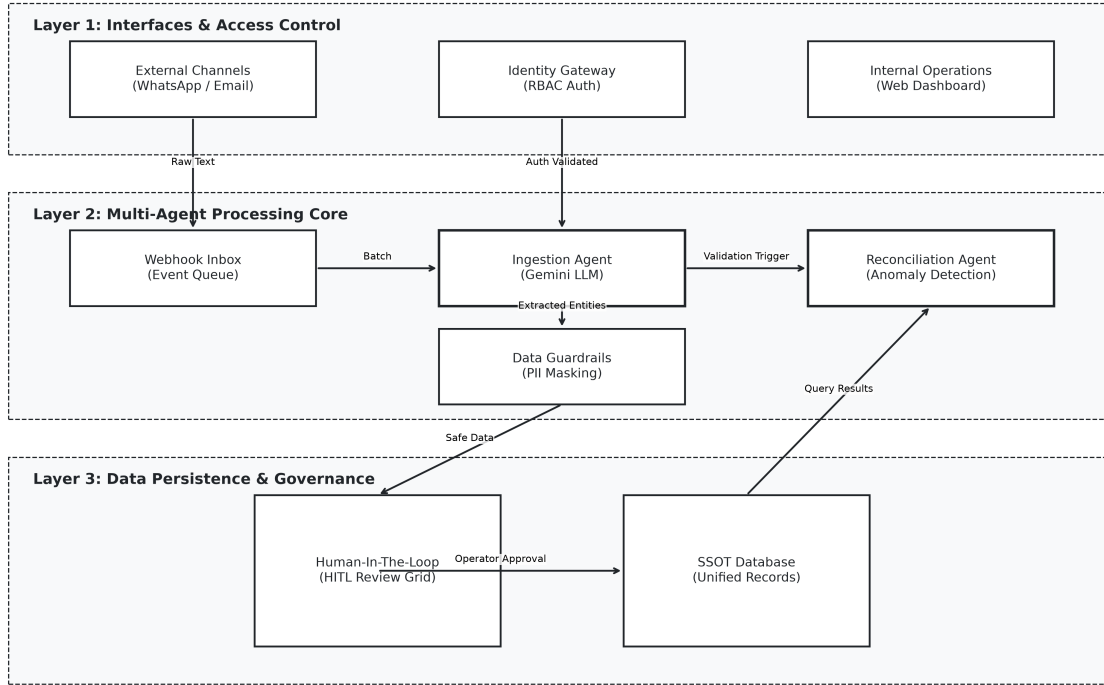


Figure 2: Proposed Multi-Agent System Architecture Flowchart

Key Enterprise Features & Security Mitigations

Deploying AI in financial workflows carries inherent risks (e.g., hallucinated refund amounts, unauthorized access, or PII leaks). Our architecture mitigates this using a **Defense-in-Depth** approach:

- **Identity Gateway & RBAC:** The application enforces dual-role authorization. Managers have full access, while Operators are restricted.
- **Data Privacy (PII Redaction):** Sensitive identifiers in unstructured messages are automatically redacted via RegEx prior to ingestion. Furthermore, Operator roles see dynamically masked UI views.
- **Data Loss Prevention (DLP):** UI-level CSS injection prevents text selection and copying of sensitive fields.
- **AI Confidence Guardrails:** The ingestion agent scores its own extractions and flags low-confidence data for mandatory human review.
- **Human-In-The-Loop (HITL) & Auditability:** The AI is strictly limited to *drafting* emails and structuring data. A human must click "Approve & Send", and every action is logged to a secure CSV for financial compliance.
- **Unified Database Explorer:** A global search interface completely breaks down the silos between Support and Finance, serving as the ultimate Single Source of Truth.

5. Measuring Success

To ensure the solution is working, we will monitor two primary metrics over the next month:

1. **Escalation Rate:** The percentage of refund requests that result in a formal escalation. **Target: 85%+ drop within 30 days.** Because 149 mismatches + 100 drops + 24 WhatsApp leaks account for virtually all operational friction, successfully deploying this SSOT proactively eliminates the root causes of the vast majority of escalations.
 2. **Tracker Discrepancy Count:** The number of tickets marked “Closed” by Support without a matching payout record in Finance. **Target: Zero.**
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6. Assumptions Made

As the brief was deliberately incomplete, I proceeded with the following assumptions to build the prototype:

- **Finance is the ultimate authority on payout amounts:** If Support quotes 5000 INR but Finance pays 4500 INR, I assumed the 500 INR deduction is legitimate (cancellation fee) rather than a Finance error, meaning the fix requires better communication to the agent rather than a refund correction.
 - **WhatsApp/Email formats are unpredictable:** I assumed agents will not follow a strict template, necessitating an LLM to dynamically extract the `agent_name`, `route`, and intent from informal natural language.
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7. Future Validations (With More Time/Access)

To further refine the SSOT and AI architecture, having more time and access would allow us to validate the following:

1. **The Source of Asynchronous Closures:** We need to investigate why Finance processed 47 tickets that Support never logged. Does the Sales or Account Management team have a backdoor process for rushing refunds? Validating this would allow us to route all departments through the AI Ingestion Agent.
2. **Finance Deduction Rule Engine:** We currently assume the 149 mismatches are due to valid policy deductions. With access to Finance SOPs, we could validate the exact calculation (e.g., 10% airline cancellation fee). We could then upgrade the AI Reconciliation Agent to proactively calculate expected payouts and set accurate expectations with the agent on day one.
3. **Raw Escalation Text Analysis:** If we had access to the raw email/call transcripts of the escalations rather than just the tracker metadata, we could validate whether "Short Payments" or "Total Non-Payments" drive higher customer churn, allowing us to dynamically prioritize the AI's processing queue based on churn risk.